

Lesson 15 Strong Induction

Fundamental Thm of Arithmetic FTA

Every integer $n > 1$ has a prime factorization

$$n = p_1 p_2 \dots p_k \rightarrow p_1, p_2, \dots, p_k: \text{primes}$$

$$12 = 2 \times 2 \times 3$$

Basis : $n = 2$

$$2 = 2 \quad \begin{array}{l} \text{trivial prime fact}^n \\ \nwarrow \text{prime} \end{array}$$

Inductive : Assume that k has a prime factⁿ : $k = p_1 p_2 \dots p_k$

For $n = k+1$, the ind hyp is not strong enough, we need to assume more.

Strong induction $S(n)$, $n \geq a$

- Basis : same as "regular" induction (sometimes may need more than one base case)
- Inductive : Assume that statement is true for $n = a, a+1, a+2, \dots, k$ (stronger ind hyp), then prove that $S(k+1)$ is true.

Back to FTA

Every integer $n > 1$ has a prime factorization
 $n = p_1 p_2 \dots p_\ell$, $p_1, p_2, \dots, p_\ell$: primes

Basis : $n = 2$

$2 = 2$ $\leftarrow$ prime trivial prime factⁿ

Inductive : Assume that $2, 3, 4, \dots, k$

have a prime factⁿ. Then for $n = k+1$,
 If $k+1$ is prime, then $k+1$ has a
 trivial prime factⁿ, itself.

If $k+1$ is not prime, then $k+1$ is
 composite, $k+1 = mr$, $1 < m, r < k+1$

By the ind hyp, m and r have prime
 factorization :

So, $m = p_1 p_2 \dots p_\ell$, $r = q_1 q_2 \dots q_j$

$k+1 = mr = p_1 p_2 \dots p_\ell q_1 q_2 \dots q_j$

primes

$k+1$ has a prime factⁿ.

Prop $b_1 = 5, b_2 = 13, b_n = 5b_{n-1} - 6b_{n-2}$
 $b_3 = 5(13) - 6(5) = 35$ $n \geq 3$

For all $n \geq 1$, $b_n = 2^n + 3^n$

Proof Basis $n=1, n=2$

$$n=1: b_1 = 5 \stackrel{?}{=} 2^1 + 3^1 \quad \checkmark$$

$$n=2: b_2 = 13 = 2^2 + 3^2 \quad \checkmark$$

Inductive: assume that $b_n = 2^n + 3^n$
for all $n=1, 2, \dots, k$ ($k \geq 2$)

Then for $n=k+1$,

$$\begin{aligned} b_{k+1} &= 5b_k - 6b_{k-1} \\ &= 5(2^k + 3^k) - 6(2^{k-1} + 3^{k-1}) \\ &= 5(2^k) + 5(3^k) - \overset{2 \cdot 3}{3}(2^k) - 2(3^k) \\ &= 2(2^k) + 3(3^k) \\ &= 2^{k+1} + 3^{k+1} \end{aligned}$$

Fibonacci #'s

$$F_1 = 1, F_2 = 1, F_n = F_{n-1} + F_{n-2} \quad (n \geq 3)$$

1, 1, 2, 3, 5, 8, 13, ...

Lemma 1 All Fib #'s are +ve.

Basis: $F_1 = F_2 = 1 > 0$

Inductive: We assume $F_1, F_2, \dots, F_k$
are +ve. Then for $n=k+1$,

$$F_{k+1} = F_k + F_{k-1} \\ > 0 + 0$$

$$F_{k+1} > 0$$

Lemma 2 $F_{n+1} \geq F_n$ for all $n \geq 1$.

For $n=1$, $F_2 \geq F_1$ because

$$1 \geq 1$$

For $n \geq 2$, $F_{n+1} = F_n + F_{n-1}$

$> F_n$ (because $F_{n-1} > 0$)
Lemma 1

Lemma 3 $F_{n+1} \leq 2F_n$

For $n=1$, $F_2 = 1 \leq 2F_1 = 2(1) = 2$

For $n \geq 2$, $F_{n+1} = F_n + F_{n-1}$

$\leq \underbrace{F_n + F_n}_{2F_n}$ (Lemma 2: $F_{n-1} \leq F_n$)

$F_{n+1} \leq 2F_n$ for all $n \geq 1$.

$$\textcircled{1} \quad 0 < F_2 < F_3 < F_4 < F_5 \dots$$

$$\textcircled{2} \quad F_{n+1} \leq 2F_n \text{ for } \forall n$$

Prop Every +ve integer can be written as a sum of distinct Fib #'s.

Basis : Base case $n=1$

$$1 = F_2 \text{ (sum of one Fib \#)}$$

Inductive : Suppose that $1, 2, \dots, k$ can be written as a sum of distinct Fib #'s. (for $k \geq 1$). Then for $n=k+1$,

If $k+1$ is a Fib #, then we are done.

If $k+1$ is not a Fib #, we can write:

$$k+1 = F_j + m$$

F_j : largest Fib # less than $k+1$

Since $0 < m < k+1$, by ind hyp, m can be written as a sum of Fib #'s.

Moreover, $m < F_j$, we prove by contradiction

$$\text{If } m \geq F_j, \quad k+1 = F_j + m \geq F_j + F_j = 2F_j \geq F_{j+1}$$

Since $k+1$ is not Fib #, $k+1 > F_{j+1} > F_j$, contradicting that F_j is the largest Fib # less than $k+1$.

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Since $m < F_j$, m as a sum of distinct
Fib #'s will not include F_j , so

$k+1 = F_j + m$ can be written
as a sum of distinct Fib #'s